



Pneumatic Controllers:-

1. Nozzle-flapper mechanism
2. Pneumatic relays

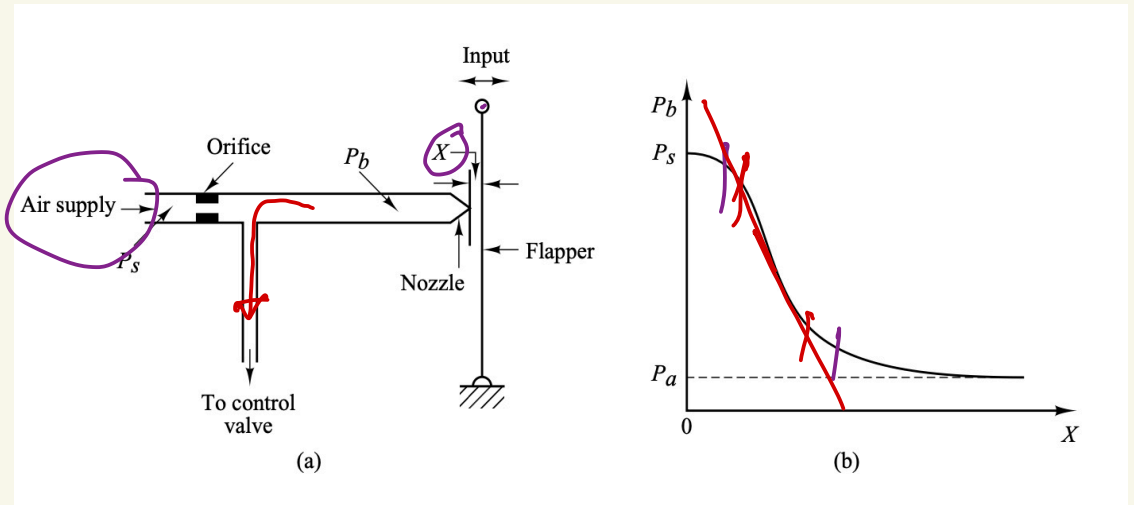


Figure 1 : Nozzle-flapper mechanism

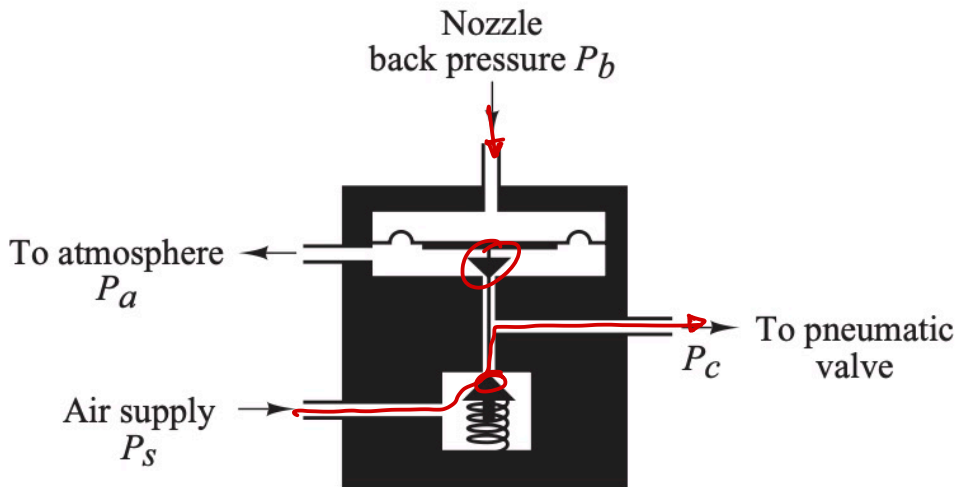
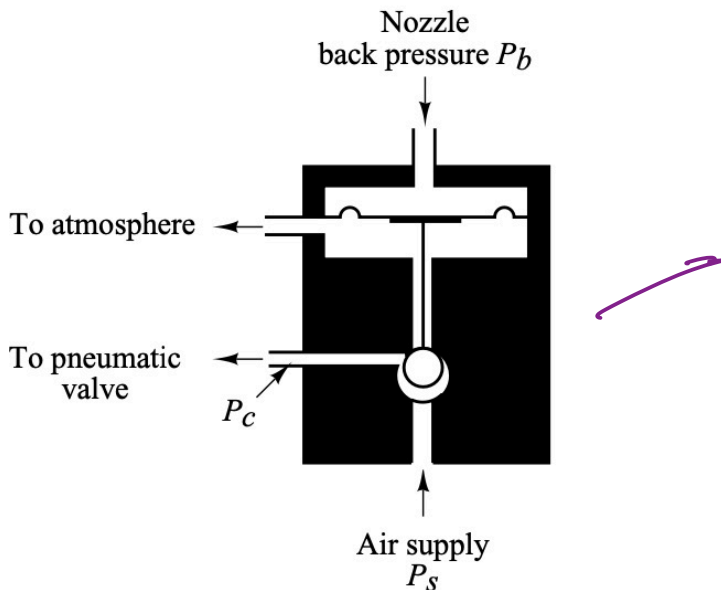
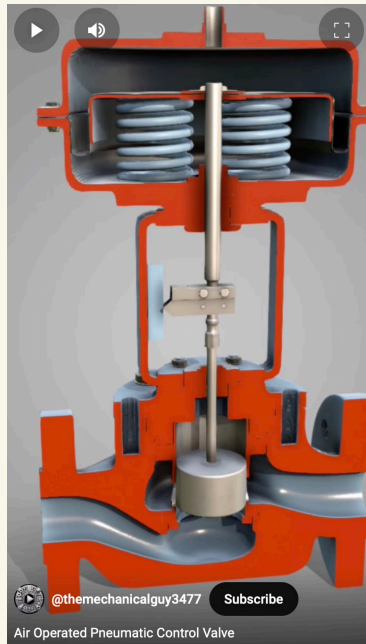


Figure 2: Pneumatic Relays



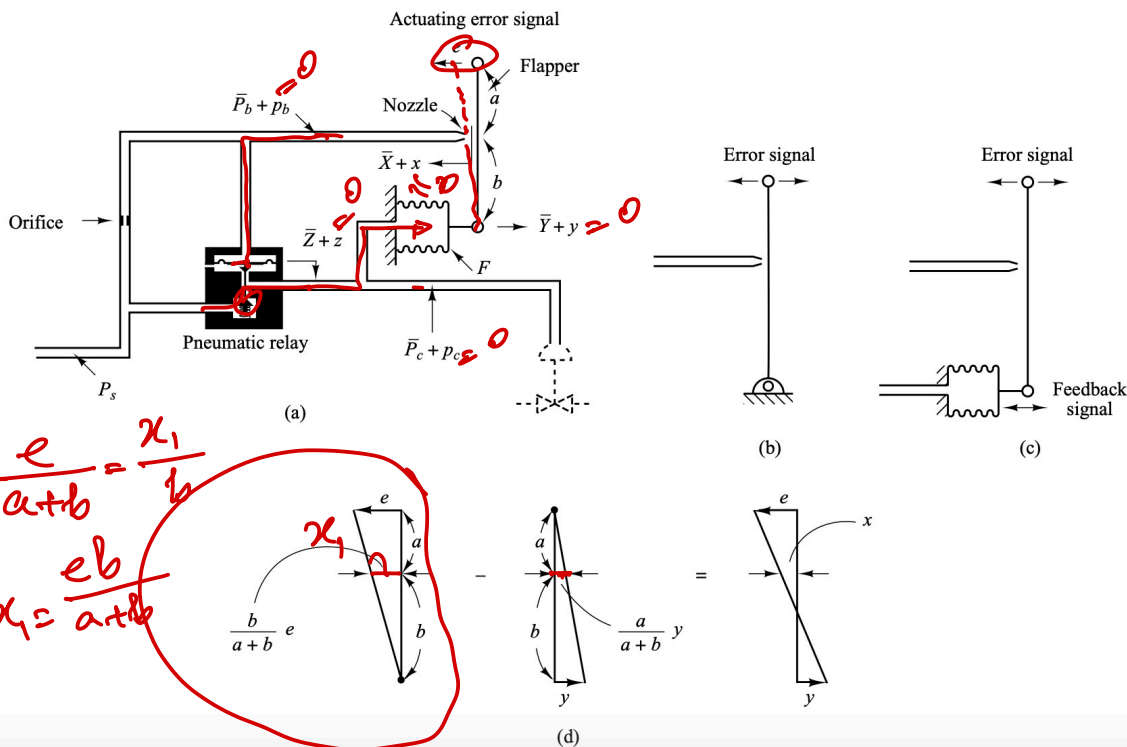


The transfer function of a PID controller can also be represented as

$$\frac{U(s)}{E(s)} = K_p \left(1 + \frac{1}{T_i s} + T_d s \right)$$

T_i = integral time

T_d = Derivative time



$$P_b = K_1 x$$

$$P_b = K_2 z \Rightarrow K_2 \cdot \frac{P_c}{K_3} = P_b \Rightarrow P_c = \frac{K_3}{K_2} P_b$$

$$P_c = K_3 z \Rightarrow \frac{P_c}{K_3} = z$$

$$\Rightarrow P_c = \frac{K_3}{K_2} \cdot K_1 x$$

$$\Rightarrow P_c = K x$$

$$A P_c = K_f y$$

$$\Rightarrow \frac{A}{K_f} \cdot P_c = y$$

$$x = \frac{b}{a+b} e - \frac{a}{a+b} y$$

$$\frac{P_c}{k} = \frac{b}{a+b} e - \frac{a}{a+b} \cdot \frac{A}{k_s} P_c$$

$$\Rightarrow \frac{P_c}{k} + \frac{a}{a+b} \cdot \frac{A}{k_s} P_c = \frac{b}{a+b} \cdot e$$

$$\Rightarrow P_c \left(\frac{1}{k} + \frac{a}{a+b} \cdot \frac{A}{k_s} \right) = \frac{b}{a+b} \cdot e$$

$$\Rightarrow P_c \cdot \frac{(a+b) \cdot k_s + k a A}{k (a+b) \cdot k_s} = \frac{b}{a+b} \cdot e$$

$$\Rightarrow \frac{P_c}{\frac{b}{a+b} \cdot e} = \frac{k}{1 + k \frac{a}{a+b} \cdot \frac{A}{k_s}}$$

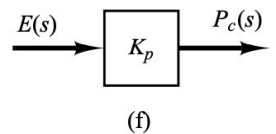
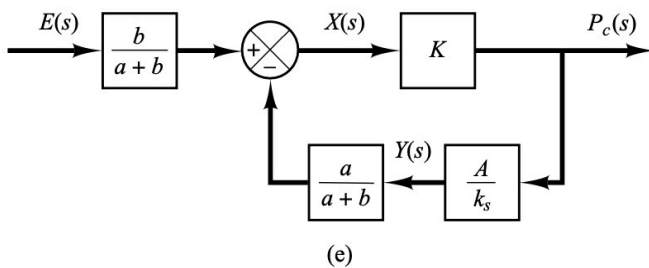
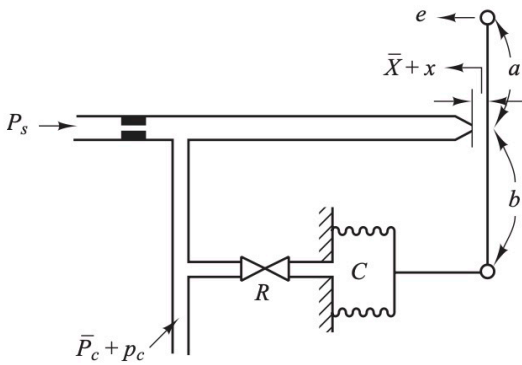
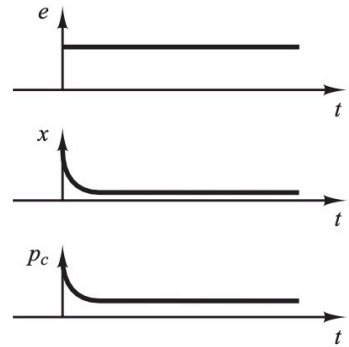


Figure: Proportional Controllers

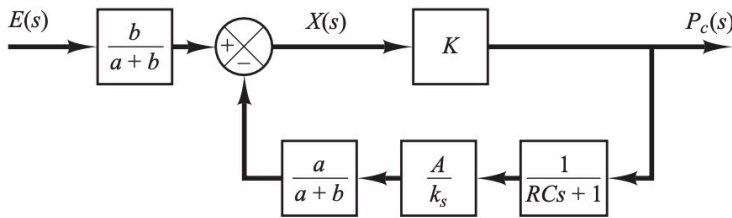
PD Controller : (proportional + Derivative)



(a)



(b)



(c)

Figure: Proportional Plus Derivative controllers. Notice the inclusion of $\frac{1}{RCs+1}$ term in the feed-back loop. The resulting equation would be

$$\frac{P_c(s)}{E(s)} = \frac{\frac{b}{a+b} K}{1 + \frac{Ka}{a+b} \cdot \frac{A}{K_s} \cdot \frac{1}{RCs+1}}$$

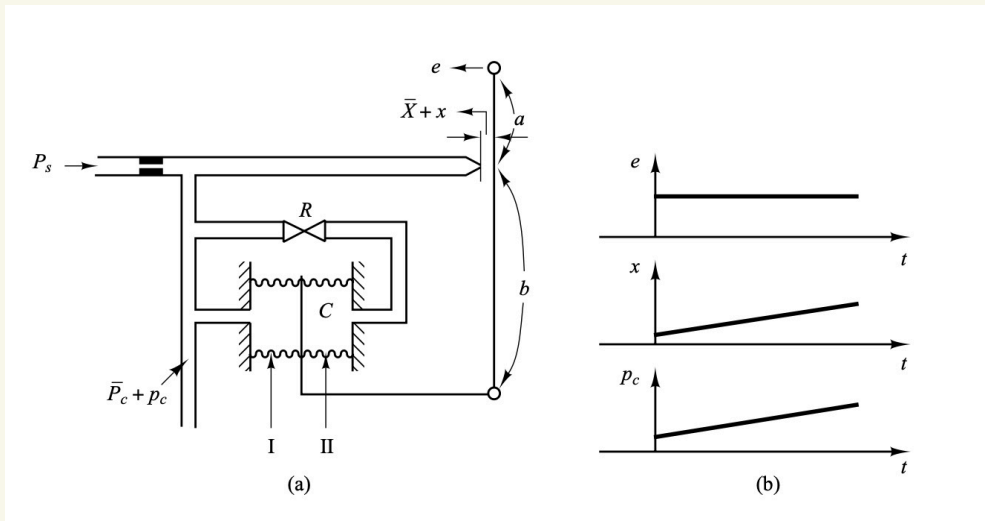
$\left| \frac{K a A}{(a+b) K_s (RCs+1)} \right|$ is made large compared to 1. In that case, the transfer function simplifies to give : (simplify)

$$\frac{P_c(s)}{E(s)} = \frac{b K_s (RCs+1)}{a A}$$

$$= K_p (1 + T_d s)$$

$$\text{where, } K_p = \frac{b K_s}{a A} \quad T_d = RC$$

Proportional plus integral Controller:
 In this case, the below is divided into two section I and II as shown in the figure.



Note that part II is connected with value R . This value creates a delayed response and the below

Π gradually attains the value of back pressure P_c . This is how the integral action is achieved in practice.

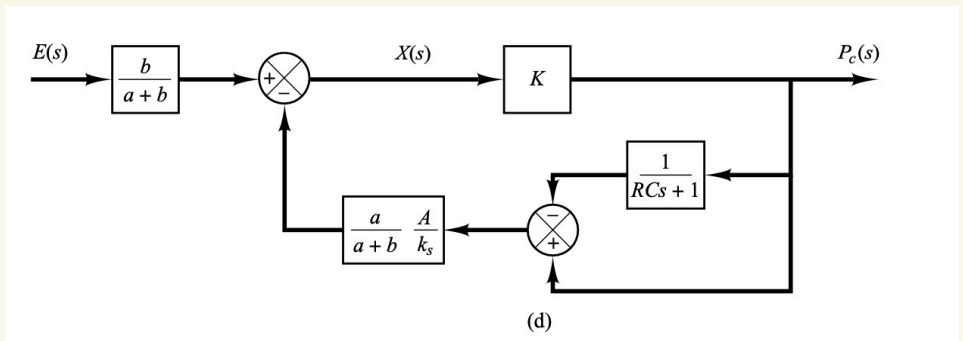


Figure: Proportional plus integral controller block diagram.

The transfer function is as follows:

$$\frac{P_c(s)}{E(s)} = \frac{\frac{b}{a+b} K}{1 + K \frac{a}{a+b} \cdot \frac{A}{k_s} \cdot \left(1 - \frac{1}{(RCs+1)}\right)}$$

The magnitude $\left| \frac{K a A R C s}{(a+b) k_s (R C s + 1)} \right|$

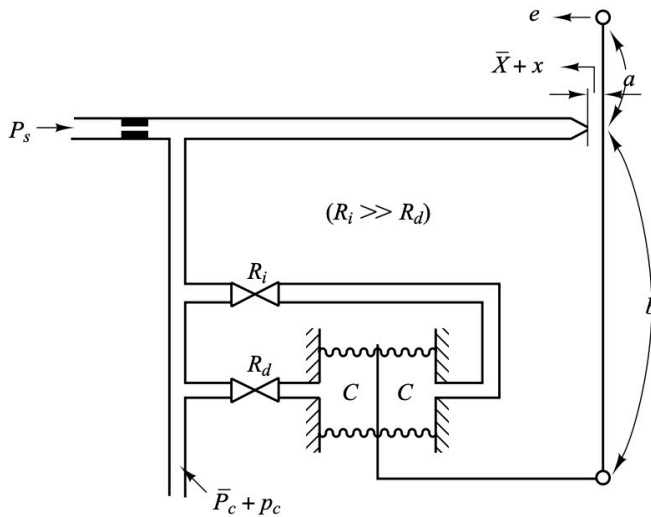
is very large compared to 1.

And the transfer function reduces to

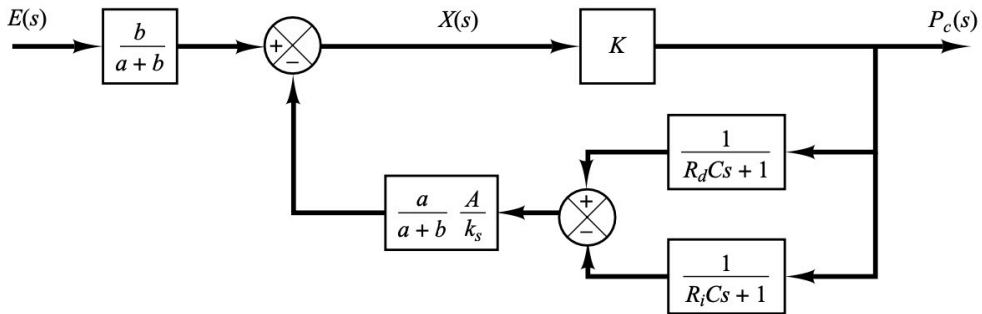
$$\frac{P_c(s)}{E(s)} = k_p \left(1 + \frac{1}{T_i s} \right)$$

$$k_p = \frac{b k_s}{a A} \quad T_i = RC$$

PID controller :



(a)



(b)

Figure: PID controller

In order to achieve the PID controller, valves are added in front of the two sections of the bellows.

The transfer function of this controller is

$$\frac{P_c(s)}{E(s)} = \frac{\frac{bk}{a+b}}{1 + \frac{ka}{a+b} \cdot \frac{A}{k_s} \cdot \frac{(R_i C - R_d C)s}{(R_d C s + 1)(R_i C s + 1)}}$$

Define $T_i = R_i C$

$T_d = R_d C$

and $\rightarrow T_i \gg T_d$

and $\rightarrow \left| \frac{kaA (\bar{T}_i - T_d)s}{(a+b)k_s (T_d s + 1)(\bar{T}_i s + 1)} \right| \gg 1$

With this the transfer function reduces to

$$\frac{P_c(s)}{E(s)} = \frac{bk_s}{aA} \cdot \frac{(T_d s + 1)(\bar{T}_i s + 1)}{(\bar{T}_i - T_d)s}$$

$$= \frac{bk_s}{aA} \cdot \frac{T_d \bar{T}_i s^2 + \bar{T}_i s + 1}{\bar{T}_i s}$$

$$\frac{P_c(s)}{E(s)} = k_p \left(1 + \frac{1}{\bar{T}_i s} + T_d s \right)$$

